

Axel Refalo

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Holding a Master's in Automation Engineering from École de Technologie Supérieure and a Master's in Mechanical Engineering from the University of Technology of Belfort-Montbéliard. I am currently pursuing a Master's thesis on the morphology of industrial robot arms, and I am looking to continue researching in robotics through a PhD

EDUCATION

École de Technologie Supérieure - ÉTS

M.Sc. in Automation Engineering, research-based

Montréal, Canada

Sept. 2024 - now

- Master's thesis on kinematic singularities and path planning for serial robot arms at CoRo
- Publication at the IEEE International Conference on Robotics and Automation
- 5th Summer School on Singularities of Mechanisms and Robotic Manipulators

Vienna - 2026

Barcelona - 2025

McGill University

Computer Science, exchange program

Montréal, Canada

Sept. 2024 - May 2025

- Applied Machine Learning, applied projects for pattern recognition and classification in Python
- Applied Robotics, controller packages on Robot Operating System with Gazebo in C++

University of Technology of Belfort-Montbéliard - UTBM

B.Eng. and M.Eng. in Mechanical Engineering, specialization in Mechatronics

Belfort, France

Sept. 2020 - Feb. 2026

- Science courses covering mechanics (control systems, continuum mechanics, finite element methods, computer-aided design and additive manufacturing), electronics (electronic design, signal processing) and computer science (LabVIEW, C++ and C# languages and calculus)

University of Palermo

Mechanical Engineering, exchange program

Palermo, Italy

Sept. 2021 - Feb. 2022

- Undergraduate courses in Italian on kinematics and dynamics of mechanisms

RESEARCH

Control and Robotics Laboratory, CoRo - ÉTS

Master's thesis

Montréal, Canada

May 2025 - now

- Comparative analysis of singularities, manipulability and cuspidality for six-revolute-joint collaborative robots from Universal Robot, Doosan, ABB and Fanuc manufacturers
- Implemented an Inverse Kinematics solver for a general six-revolute-joint serial manipulator
- Publication: Refalo, A., Bonev, I., and Gosselin, C. "Singularity Analysis of ABB's GoFa 5 Robot Arm," accepted at International Conference on Robotics and Automation 2026

Laboratory of Bioresources - McGill

Research project

Montréal, Canada

Jan. 2025 - now

- Robotic platform for collection of plant growth data and to model spatiotemporal growth patterns, design and control from scratch
- In preparation: Debbagh, M., Refalo, A., et al. "An Affordable and Open-Source Gantry Framework for Capturing High Resolution Spatiotemporal Plant Growth Data"

Materials Science Laboratory - ÉTS

Research project

Montréal, Canada

Dec. 2024 - May 2025

- Crack propagation analysis for different concrete mixes, focusing on computed tomography reconstruction and filtering of synchrotron concrete scans

Laboratory of Space Technologies - ÉTS <i>Research project</i>	Montréal, Canada Sept. 2024 - Dec. 2024
- Kalman filter for position estimation using signals of opportunity from Low Earth Orbit satellite constellations	
Canon Production Printing - R&D <i>Internship</i>	Eindhoven, Netherlands Sept. 2023 - Feb. 2024
- Two Cartesian robots development: (I) for inspecting static microscopic print nozzle defects and contamination, and (II) for inspecting dynamic behavior of the ink on the coating surface	

PROJECTS

Robotics	
- Two-cable-driven parallel plotter for spray painting	2026
- Three-degree-of-freedom manipulator with parallelogram mechanism	2024
- Wide planar plotter driven differentially	2022
- Other robotics projects: simple robot arm, modular robot chain, hovercrafts, and electric skateboards	
Computer Aided Design	
- All the previous robotics project mechanical and electronic designed and built	
- Replication of a Swiss watch mechanism, design of parametric gears with a cycloidal profile	2023
Programming	
- All the previous robotics projects, coded mostly in C++ or Python	
- Unix Home lab server built and accessed remotely	2025
- Simple Eulerian fluid simulator	2023
- Algorithms to solve complex grid puzzles: sudoku, kakuro, killer sudoku, suguru using backtracking	2023
- Mechanical truss simulator using finite element methods	2022

SKILLS

Languages	French (Native), English (Bilingual - IELTS), Italian (Bilingual - IB), Spanish (Beginner)
Software	<i>Robotics:</i> Robot Operating System, Gazebo, RoboDK, RobotStudio and Roboguide <i>Programming:</i> Python, C++, C#, LabVIEW, Unix shell, LaTeX and Visual Studio Code <i>Math:</i> MATLAB, SciPy, Wolfram, Maple and GeoGebra
Mechanics	<i>Computer Aided Design:</i> CATIA V5, Fusion 360 and AutoCAD <i>Physical modeling:</i> Simulink, Amesim, Ansys, OrCAD, Abaqus and DragonFly
Electronics	Controller board (Raspberry Pis and Arduinos) motors (stepper and brushless), batteries and power converters, and usual electronic components
Manufacturing	Additive printing, laser cutter, electronic soldering and hand tools

INTERESTS

<i>Driven by challenge and curiosity · Travel across Canada, Netherlands, Italy and Spain · Arts: literature, cinema, documentary and drawing · Sport: chess, hiking and swimming · And a deep passion for applying science through collaborative engineering challenges, including:</i>	
- Employed as FabLab manager, design department of ÉTS	2025
- CrunchTime hackathon hosted by UTBM: participant (2023) and electronics manager (2024)	
- Crunch industry camp hackathon hosted by UTBM: finalist	2022